



NATIONAL UNIVERSITY OF TECHNOLOGY

(Department of Computer Engineering)

PBL

Course Title	: Discrete Structures	Course code	: MATH3901
Batch Number	: CEN-24	Assign Date	: 31-05-2025
Total Marks	: 10	Due Date	: 16-06-2025
FM Name	: Dr. Muhammad Sajjad	Semester	: Spring 2025
Student Name	: _____	Student ID	: _____

Note: Carefully write up solutions. Show all of your work. Minimal to no credit will be given for solutions without appropriate justification.

Redesigning a 4×4 Nonlinear Component of Block Cipher based on Graph Theory, Galois Field and Discrete Structures Operations

Apply the concepts of graph theory, Galois field and discrete mathematics to redesign the nonlinear component of a 4×4 block cipher through the following steps:

1. Graph: Draw a graph with 4 vertices and at least 4 edges.
2. Matrix Representation: Construct the corresponding matrix M for the graph drawn in Step 1. (Note: A different matrix M assigned to each group.)
3. Galois Field Generation: List all the elements of the Galois Field $GF(2^4)$.
4. Affine Transformation: Apply the affine transformation defined by:

$$f(x_i) = Mx_i^T \oplus b^T,$$

where:

- M is the matrix from Step 2
 - x_i are the elements of $GF(2^4)$
 - b is any non-zero fixed value from $GF(2^4)$
 - $\oplus$ denotes the bitwise XOR operation
5. S-box Construction: Arrange the output values of the affine transformation into a 4×4 matrix, forming a 4×4 nonlinear component (S-box) for the block cipher. This component can be utilized in applications such as the Internet of Things (IoT) and Cybersecurity.
 6. Implementation: Write a program to implement the above steps using MATLAB, Python, C++, or any other suitable programming language.